7.5.5 Darcy's formula for loss of head in pipes in terms of velocity

Head loss due to friction (hf) = $\frac{flv^2}{2gd}$

Where,

f = coefficient of friction

l = length of pipe

v = velocity of water in the pipe

g = acceleration due to gravity (9.81 m/s^2)

d = diameter of the pipe

7.5.6 Darcy's formula for loss of head in terms of discharge

Head loss due to friction (hf) = $\frac{flQ^2}{12d^5}$

Where,

f = coefficient of friction

l = length of pipe

Q = discharge of water from the pipe

d = diameter of the pipe

7.5.7 Problems

Example – 1

Water flows through a pipe of 100 mm diameter and 120 m long with a velocity of 3 m/sec. Find the loss of head due to friction by using Darcy's formula. Take the friction factor as 0.002.

Given data:

Diameter of pipe (d) = 100 mm = 0.10 m

Length of pipe (l) = 120 m

Velocity of flow (v) = 3 m/sec

Friction factor (f) = 0.002

To find out:

Loss of head due to friction (h_f)

Solution:

Loss of head due to friction

$$(h_f) = \frac{flv^2}{2gd}$$

$$= \frac{0.002 \times 120 \times 3^2}{2 \times 9.81 \times 0.10} = 1.10\,m$$

Result:

Loss of head, h_f = 1.10 m

Example – 2

A pipe line, 300 mm diameter and 400 m long connects two reservoirs. The discharge through the pipe is 191 lps. Taking the friction factor as 0.002, determine the loss of head due to friction.

Given data:

Diameter of the pipe (d) = 300 mm = 0.30 m

Length of the pipe (l) = 400 m

Discharge of water (Q) = 191 lps = 0.191 m^3/sec

Friction factor (f) = 0.002

To determine:

Loss of head due to friction, h_f.

Solution:

Loss of head due to friction

$$(h_f) = \frac{flQ^2}{12d^5}$$

$$= \frac{0.002 \times 400 \times 0.191^2}{12 \times 0.30^5}$$

$$= 1.00 \text{ m}$$

Result:

Loss of head, hf = 1.00 m.

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